

Freight Rate Prediction: Validation Report

1. Approach

Model: LightGBM regression on $\log(\text{posted_rate})$, averaged over 3 seeds, predictions converted back to dollars.
Inputs: distance, log distance, $|\text{weight}|$ + missing flag, equipment, pickup/delivery city and coordinates, straight-line distance, road/straight-line ratio, day of week. A median \$/mile-by-distance baseline is used as the reference point.

2. Data quality and handling

- **Negative weights** (292 dev, 145 validation rows): the absolute values match the positive distribution (5k-47.5k lb), so they are treated as sign errors and flipped.
- **Missing weight** (300 dev, 165 validation): left as NaN with a missing flag; LightGBM handles it natively.
- **Missing market_index** (374 dev rows): irrelevant to the final model, which does not use it.
- **Corrupted targets**: ~1.41% of labeled rows (677) have a rate roughly 3-5x or 0.2-0.4x the expected value for the lane, spread evenly across months and unrelated to any feature. They are flagged with an out-of-fold Huber model ($|\log \text{residual}| > 0.30$) and removed from training only. Evaluation rows are never removed.
- **Short-haul distances**: distances appear floored near 70 miles, so a few short lanes have a distance far above straight-line distance. The rate follows the listed distance, so these were kept.

3. Split and validation

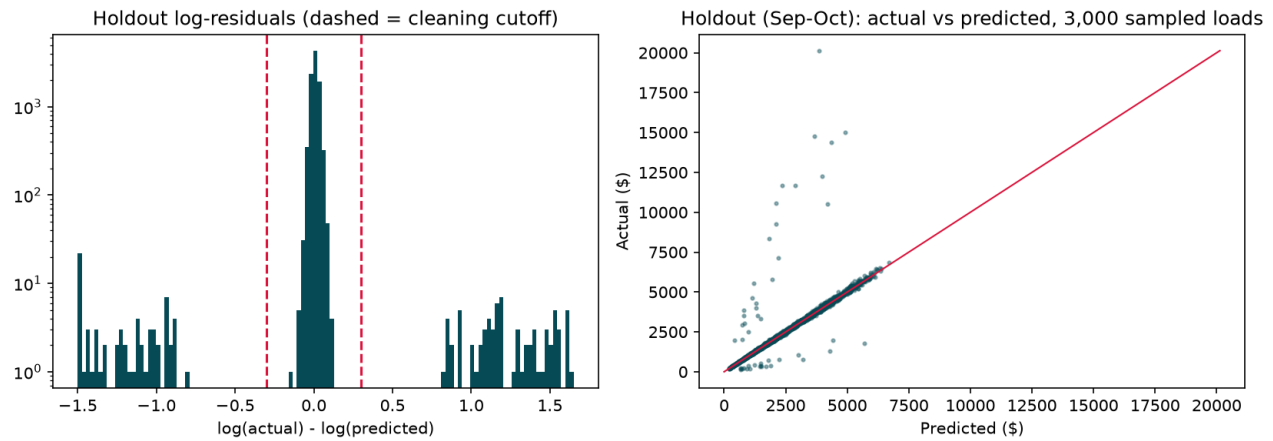
The labeled data covers Jan-Oct 2025 and the 12,000 validation loads cover Nov-Dec 2025, so the task is a forecast. I used expanding-window, time-based folds instead of a random split. Folds 1-2 were used for hyperparameter selection; fold 3 (train Jan-Aug, test Sep-Oct) is the final holdout, evaluated once after tuning.

fold	test_window	MAE	RMSE	MAPE_%	MedAPE_%
1	2025-07-01..2025-08-31	104.25	623.30	4.46	1.90
2	2025-08-01..2025-09-30	93.17	613.26	4.09	1.52
3	2025-09-01..2025-10-31	98.59	630.60	4.37	1.56

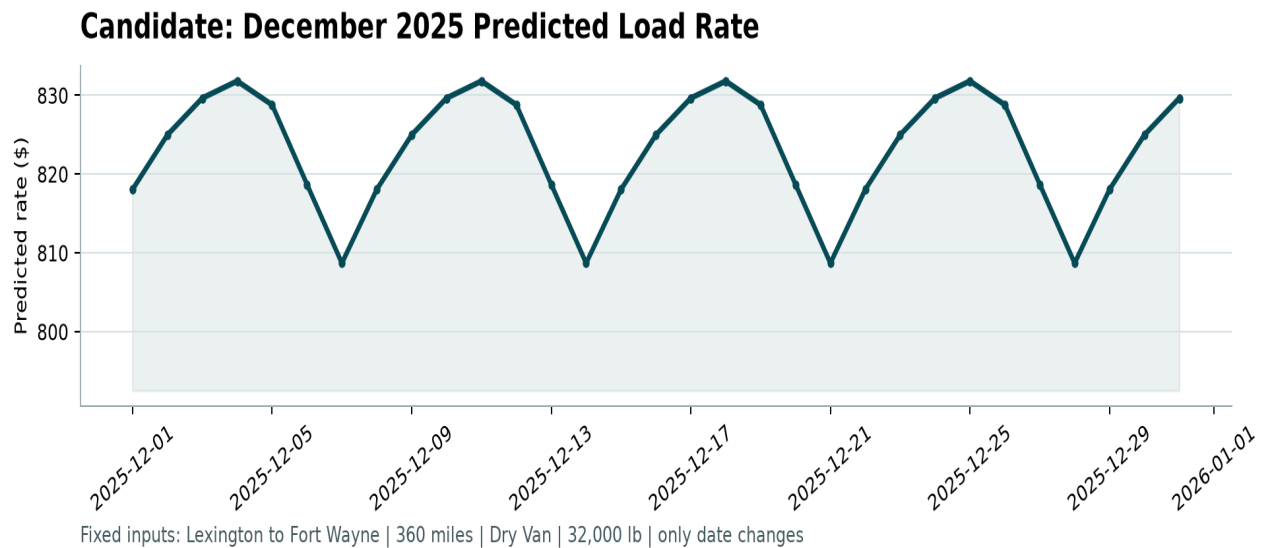
Final holdout (Sep-Oct) and ablations

model	MAE	RMSE	MAPE_%	MedAPE_%
Baseline: median \$/mile by distance bucket x distance	191.56	656.02	8.36	5.44
LightGBM, no target cleaning	121.26	635.69	5.37	2.25
LightGBM + target cleaning (FINAL)	98.59	630.60	4.37	1.56
+ market_index & quote_signal as features	149.42	644.77	6.36	3.89
(random 80/20 split) final features	116.80	622.48	4.96	2.45
(random 80/20 split) + market_index & quote_signal	91.61	617.37	3.93	1.54

Findings. (a) Cleaning corrupted targets cut holdout MAPE from 5.4% to 4.4%. (b) RMSE stays near 630 because the holdout itself contains ~1.5% corrupted rows that no model should predict; on non-corrupted rows MAPE is about 1.9%. (c) market_index and quote_signal improve a random split but hurt the forward split, so they are excluded. (d) The random split is a less reliable guide to forward performance, which is why the time-based split was used.



4. Fixed December chart (produced by score.py)



The chart shows a weekly cycle (lowest on Sundays, highest on Thursdays) of roughly \$808-832 for the fixed Lexington to Fort Wayne dry-van load. Because `market_index` is not an input, the model has no month-level trend beyond the day-of-week pattern, which is why the pattern repeats each week.